OCQKUDSIN

A simple mathematical guide to the sine-based Krylov method

Hamiltonian scaling → sine operator → Krylov subspace → energy estimate

Purpose. Build a small set of useful states from a reference state, then use that set to estimate low energies of a molecular Hamiltonian. The implementation in `ocqkudsin.py` uses the Hermitian operator $X_\epsilon = \sin(\epsilon \bar{H})$ and circuit estimates of its moments.

1. Scale the Hamiltonian and define the operator

Write the physical qubit Hamiltonian H as a sum of Pauli strings P_ℓ,

$$H = \sum_\ell h_\ell P_\ell, \qquad \lambda = \sum_\ell |h_\ell|, \qquad \bar{H} = H/\lambda.$$

Here ℓ labels Pauli terms, h_ℓ are their real coefficients, λ is the coefficient one-norm, and $\bar{H}$ is the dimensionless scaled Hamiltonian. Since each P_ℓ has operator norm one, $\|\bar{H}\| \le 1$. The reference $|\Phi_0\rangle$ is the Hartree–Fock state. For a real, dimensionless parameter ϵ, define

$$U_\epsilon = i \exp(-i\epsilon \bar{H}),$$
$$X_\epsilon = \frac{U_\epsilon + U_\epsilon^\dagger}{2} = \sin(\epsilon \bar{H}).$$

Here $i = \sqrt{-1}$ and $\dagger$ denotes conjugate transpose. The factor i is a global phase on the time-evolution unitary; it makes its Hermitian part the sine rather than the cosine. X_ϵ is Hermitian and has spectrum in $[-1, 1]$.

2. Generate the Krylov states

The index $n = 0, 1, \ldots$ counts Lanczos steps; $|q_n\rangle$ is the normalized state at step n. Start with $|q_0\rangle = |\Phi_0\rangle$, set the dummy state $|q_{-1}\rangle = 0$ and $\beta_0 = 0$. The three-term Lanczos relation is

$$\alpha_n = \langle q_n | X_\epsilon | q_n \rangle,$$
$$r_n = X_\epsilon |q_n\rangle - \alpha_n |q_n\rangle - \beta_n |q_{n-1}\rangle,$$
$$\beta_{n+1} = \|r_n\|, \qquad |q_{n+1}\rangle = r_n/\beta_{n+1}.$$

Here α_n is the diagonal moment, r_n is the residual vector, and β_n and β_{n+1} are nonnegative recurrence coefficients; $\|\cdot\|$ is the state-vector norm. Equivalently, $X_\epsilon |q_n\rangle = \beta_{n+1}|q_{n+1}\rangle + \alpha_n |q_n\rangle + \beta_n |q_{n-1}\rangle$. Stop when the requested dimension is reached or β_{n+1} is near zero.

Moments measured by the circuits

For a normalized state $|q_n\rangle$, the implementation estimates

$$\alpha_n = \mathrm{Re}\langle q_n | U_\epsilon | q_n \rangle, \qquad \mu_{2,n} = \langle q_n | X_\epsilon^2 | q_n \rangle = \frac{1 + \mathrm{Re}\langle q_n | U_\epsilon^2 | q_n \rangle}{2}.$$

Here $\mu_{2,n}$ is the second moment of X_ϵ, and Re means real part. The notation U_ϵ^2 means $U_\epsilon U_\epsilon$. Then $\beta_{n+1} = \sqrt{\mu_{2,n} - \alpha_n^2 - \beta_n^2}$. Hadamard-test circuits estimate the real overlaps. At $\epsilon = 0$, $X_0 = 0$, so the Krylov space contains only $|q_0\rangle$.

3. Generate each state with Chebyshev coefficients

Chebyshev polynomials provide the basis used to represent and evaluate each Lanczos state; they are not a second batch of states computed after the $|q_n\rangle$. Let T_m be the first-kind Chebyshev polynomial of degree $m = 0, 1, \ldots$, with scalar argument $x \in [-1, 1]$:

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x), \qquad |v_m\rangle = T_m(X_\epsilon)|\Phi_0\rangle.$$

The vectors $|v_m\rangle$ are the Chebyshev states. Write $|q_n\rangle = \sum_m c_{n,m}|v_m\rangle$, where $c_{n,m}$ are scalar expansion coefficients. Set $c_{0,0} = 1$, $c_{-1,m} = 0$, and treat coefficients outside their stored range as zero. Multiplication by X_ϵ gives the intermediate coefficients $d_{n,m}$:

$$d_{n,0} = \tfrac{1}{2}c_{n,1}, \qquad d_{n,1} = c_{n,0} + \tfrac{1}{2}c_{n,2}, \qquad d_{n,m} = \tfrac{1}{2}(c_{n,m-1} + c_{n,m+1}) \quad (m \geq 2).$$

The Lanczos coefficient update is

$$c_{n+1,m} = \frac{d_{n,m} - \alpha_n c_{n,m} - \beta_n c_{n-1,m}}{\beta_{n+1}}.$$

Operationally, initialize $|q_0\rangle$ with $c_{0,0} = 1$. At each step, measure moments for $|q_n\rangle$, calculate β_{n+1}, update the coefficients $c_{n+1,m}$, then use Clenshaw's recurrence below to evaluate and normalize $|q_{n+1}\rangle$. The implementation does not build the $|v_m\rangle$ individually to generate these states; it returns them separately only for inspection.

For $|q_n\rangle = \sum_{m=0}^{m_{\max}} c_{n,m}T_m(X_\epsilon)|\Phi_0\rangle$, $m_{\max}$ is the largest stored degree. Initialize the auxiliary vectors $|b_{m_{\max}+1}\rangle = |b_{m_{\max}+2}\rangle = 0$ and work downward:

$$|b_m\rangle = 2X_\epsilon|b_{m+1}\rangle - |b_{m+2}\rangle + c_{n,m}|\Phi_0\rangle \quad (m = m_{\max}, \ldots, 1),$$

$$|q_n^{\text{raw}}\rangle = X_\epsilon|b_1\rangle - |b_2\rangle + c_{n,0}|\Phi_0\rangle, \qquad |q_n\rangle = |q_n^{\text{raw}}\rangle / \|q_n^{\text{raw}}\|.$$

The vectors $|b_m\rangle$ are temporary Clenshaw work vectors; $|q_n^{\text{raw}}\rangle$ is the unnormalized state.

4. Project onto the generated subspace

For retained-state indices i, j, measure the overlap matrix S and dimensionless projected Hamiltonian matrix M:

$$S_{ij} = \langle q_i|q_j\rangle, \qquad M_{ij} = \langle q_i|\bar{H}|q_j\rangle.$$

In exact arithmetic the Lanczos states are orthonormal, so $S = I$, where I is the identity matrix. The implementation measures S as an orthogonality diagnostic and diagonalizes M for its dimensionless Ritz eigenvalues $\bar{E}_j$ (projected energy estimates). Convert back to physical energy units with

$$\boxed{E_j = \lambda\,\bar{E}_j \quad \text{(Hartree if } H \text{ is in Hartree).}}$$

Here E_j is a physical Ritz energy. Diagonal entries are expectation values; ancilla interference circuits estimate off-diagonal real and imaginary parts. The lowest projected eigenvalue is the ground-state energy estimate.

In one line: scale H, form $X_\epsilon = \sin(\epsilon\bar{H})$, grow its Lanczos space using two measured moments per state, evaluate states through Chebyshev coefficients, and project H to obtain energies.

Source: implementation details and conventions are taken from `ocqkudsin.py`. Exact dense evolution and generic state preparation in this implementation target small systems.